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Smart bag to measure urine output via catheter

Abstract

Disclosed herein is a urine collection bag, system, and methods directed to automated measurement of a quantity of urine. The urine collection system can include the urine collection bag, a catheter, and flexible drainage tubing. The urine collection bag can include a collection area, force sensors, and circuitry configured to determine the volume of urine in the collection area based on pressure or weight measured by the force sensors, and the specific gravity of the urine. The catheter may include a small female external catheter (FEC) with an opening on a top side, a wicking catheter with a wicking area on a top side, a finger-mountable catheter, or a male external catheter (MEC). The tubing may be secured to a patient's leg with a stabilization device or a fabric strap. The catheter can remain on a patient while the patient stands or walks.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3661143	12/1971	Henkin	N/A	N/A
3781920	12/1973	Browne et al.	N/A	N/A
3851650	12/1973	Darling	N/A	N/A
3919455	12/1974	Sigdell et al.	N/A	N/A
4276889	12/1980	Kuntz et al.	N/A	N/A
4286590	12/1980	Murase	N/A	N/A
4291692	12/1980	Bowman et al.	N/A	N/A
4296749	12/1980	Pontifex	N/A	N/A
4305405	12/1980	Meisch	N/A	N/A
4312352	12/1981	Meisch et al.	N/A	N/A
4343316	12/1981	Jespersen	N/A	N/A
4443219	12/1983	Meisch et al.	N/A	N/A
4448207	12/1983	Parrish	N/A	N/A
4509366	12/1984	Matsushita et al.	N/A	N/A
4532936	12/1984	LeVeen et al.	N/A	N/A
4658834	12/1986	Blankenship et al.	N/A	N/A
4712567	12/1986	Gille	600/580	A61B 5/14507
4723950	12/1987	Lee	N/A	N/A
4834706	12/1988	Beck et al.	N/A	N/A
4850375	12/1988	Rosenberg	N/A	N/A
4889532	12/1988	Metz et al.	N/A	N/A
5002541	12/1990	Conkling et al.	N/A	N/A
5146637	12/1991	Bressler et al.	N/A	N/A
5409014	12/1994	Napoli et al.	N/A	N/A
5586085	12/1995	Lichte	N/A	N/A
5725515	12/1997	Propp	N/A	N/A
5733319	12/1997	Neilson et al.	N/A	N/A
5738656	12/1997	Wagner	N/A	N/A
5747824	12/1997	Jung et al.	N/A	N/A
5769087	12/1997	Westphal et al.	N/A	N/A

5807278	12/1997	McRae	N/A	N/A
5823972	12/1997	McRae	N/A	N/A
5891051	12/1998	Han et al.	N/A	N/A
5911786	12/1998	Nielsen et al.	N/A	N/A
6129684	12/1999	Sippel et al.	N/A	N/A
6132407	12/1999	Genese et al.	N/A	N/A
6250152	12/2000	Klein et al.	N/A	N/A
6256532	12/2000	Cha	N/A	N/A
6261254	12/2000	Baron et al.	N/A	N/A
6434418	12/2001	Neal et al.	N/A	N/A
6579247	12/2002	Abramovitch et al.	N/A	N/A
6592612	12/2002	Samson et al.	N/A	N/A
6709420	12/2003	Lincoln et al.	N/A	N/A
6716200	12/2003	Bracken et al.	N/A	N/A
7011634	12/2005	Paasch et al.	N/A	N/A
7161484	12/2006	Tsoukalis	N/A	N/A
7211037	12/2006	Briggs et al.	N/A	N/A
7437945	12/2007	Feller	N/A	N/A
7442754	12/2007	Tepper et al.	N/A	N/A
7739907	12/2009	Boiarski	N/A	N/A
7871385	12/2010	Levinson	N/A	N/A
7931630	12/2010	Nishtala et al.	N/A	N/A
7976533	12/2010	Larsson	N/A	N/A
7998126	12/2010	Fernandez	N/A	N/A
8295933	12/2011	Gerber et al.	N/A	N/A
8328733	12/2011	Forte et al.	N/A	N/A
8328734	12/2011	Salvadori et al.	N/A	N/A
8337476	12/2011	Greenwald et al.	N/A	N/A
8374688	12/2012	Libbus et al.	N/A	N/A
8403884	12/2012	Nishtala	N/A	N/A
8471231	12/2012	Paz	N/A	N/A
8663128	12/2013	Paz et al.	N/A	N/A
8773259	12/2013	Judy et al.	N/A	N/A
8790277	12/2013	Elliott et al.	N/A	N/A
8790320	12/2013	Christensen	N/A	N/A
8790577	12/2013	Mizumoto et al.	N/A	N/A
8813551	12/2013	Boiarski	N/A	N/A
8827924	12/2013	Paz et al.	N/A	N/A
8832558	12/2013	Cardarelli et al.	N/A	N/A
8900196	12/2013	Andino	N/A	N/A
9045887	12/2014	O'Malley	N/A	N/A
9050046	12/2014	Elliott et al.	N/A	N/A
9074920	12/2014	Mendels et al.	N/A	N/A
9216242	12/2014	Nishtala et al.	N/A	N/A
9480821	12/2015	Ciccone et al.	N/A	N/A
9592034	12/2016	Hall et al.	N/A	N/A
9642987	12/2016	Bierman et al.	N/A	N/A
9731097	12/2016	Andino	N/A	A61M 25/02
9895095	12/2017	Chen	N/A	N/A
9962516	12/2017	Lampotang et al.	N/A	N/A

10071202	12/2017	Handler	N/A	N/A
10182747	12/2018	Charlez et al.	N/A	N/A
10245008	12/2018	Paige	N/A	N/A
10362981	12/2018	Paz et al.	N/A	N/A
10383606	12/2018	McCord et al.	N/A	N/A
10448875	12/2018	Holt et al.	N/A	N/A
10799386	12/2019	Harrison, Sr.	N/A	A61F 5/441
10881778	12/2020	Scarpaci et al.	N/A	N/A
11540760	12/2022	Guillemette	N/A	N/A
11703365	12/2022	Tourchak et al.	N/A	N/A
12109353	12/2023	Cheng et al.	N/A	N/A
2001/0056226	12/2000	Zodnik et al.	N/A	N/A
2002/0016719	12/2001	Nemeth et al.	N/A	N/A
2002/0161314	12/2001	Sarajarvi	N/A	N/A
2002/0193760	12/2001	Thompson	N/A	N/A
2003/0000303	12/2002	Livingston et al.	N/A	N/A
2003/0163183	12/2002	Carson	N/A	N/A
2003/0163287	12/2002	Vock et al.	N/A	N/A
2004/0267086	12/2003	Anstadt et al.	N/A	N/A
2005/0020958	12/2004	Paolini et al.	N/A	N/A
2005/0065583	12/2004	Voorhees et al.	N/A	N/A
2005/0172712	12/2004	Nyce	N/A	N/A
2005/0247121	12/2004	Pelster	N/A	N/A
2006/0065713	12/2005	Kingery	N/A	N/A
2006/0100743	12/2005	Townsend et al.	N/A	N/A
2006/0253091	12/2005	Vernon	604/323	A61F 5/44
2007/0010797	12/2006	Nishtala et al.	N/A	N/A
2007/0078444	12/2006	Larsson	N/A	N/A
2007/0106177	12/2006	Hama	600/573	G01G 17/04
2007/0145137	12/2006	Mrowiec	N/A	N/A
2007/0225668	12/2006	Otto	N/A	N/A
2007/0252714	12/2006	Rondoni et al.	N/A	N/A
2008/0027409	12/2007	Rudko et al.	N/A	N/A
2008/0217391	12/2007	Roof et al.	N/A	N/A
2008/0312550	12/2007	Nishtala et al.	N/A	N/A
2008/0312556	12/2007	Dijkman	N/A	N/A
2009/0056020	12/2008	Caminade et al.	N/A	N/A
2009/0099629	12/2008	Carson et al.	N/A	N/A
2009/0157430	12/2008	Rule et al.	N/A	N/A
2009/0287170	12/2008	Otto	N/A	N/A
2009/0315684	12/2008	Sacco et al.	N/A	N/A
2010/0064426	12/2009	Chikara Imamura	N/A	N/A
2010/0094204	12/2009	Nishtala	N/A	N/A
2010/0130949	12/2009	Garcia	N/A	N/A
2010/0137743	12/2009	Nishtala et al.	N/A	N/A
2011/0113540	12/2010	Plate et al.	N/A	N/A
2011/0120219	12/2010	Barlesi et al.	N/A	N/A
2011/0178425	12/2010	Nishtala et al.	N/A	N/A
2011/0224636	12/2010	Keisic	N/A	N/A
2011/0230824	12/2010	Salinas et al.	N/A	N/A

2011/0238042	12/2010	Davis et al.	N/A	N/A
2011/0251572	12/2010	Nishtala et al.	N/A	N/A
2011/0263952	12/2010	Bergman et al.	N/A	N/A
2012/0029408	12/2011	Beaudin	N/A	N/A
2012/0035496	12/2011	Denison et al.	N/A	N/A
2012/0059286	12/2011	Hastings et al.	N/A	N/A
2012/0078137	12/2011	Mendels et al.	N/A	N/A
2012/0078235	12/2011	Martin et al.	N/A	N/A
2012/0095304	12/2011	Biondi	N/A	N/A
2012/0109008	12/2011	Charlez et al.	N/A	N/A
2012/0118650	12/2011	Gill	N/A	N/A
2012/0123233	12/2011	Cohen	N/A	N/A
2012/0127103	12/2011	Qualey et al.	N/A	N/A
2012/0226196	12/2011	DiMino et al.	N/A	N/A
2012/0234434	12/2011	Woodruff et al.	N/A	N/A
2012/0302917	12/2011	Fitzgerald et al.	N/A	N/A
2012/0323144	12/2011	Coston et al.	N/A	N/A
2012/0323502	12/2011	Tanoura et al.	N/A	N/A
2013/0066166	12/2012	Burnett et al.	N/A	N/A
2013/0109927	12/2012	Menzel	N/A	N/A
2013/0109928	12/2012	Menzel	N/A	N/A
2013/0131610	12/2012	Dewaele et al.	N/A	N/A
2013/0218106	12/2012	Coston et al.	N/A	N/A
2013/0245498	12/2012	Delaney et al.	N/A	N/A
2013/0267871	12/2012	Delaney et al.	N/A	N/A
2014/0039348	12/2013	Bullington et al.	N/A	N/A
2014/0155781	12/2013	Bullington et al.	N/A	N/A
2014/0155782	12/2013	Bullington et al.	N/A	N/A
2014/0159921	12/2013	Qualey et al.	N/A	N/A
2014/0187666	12/2013	Aizenberg et al.	N/A	N/A
2014/0207085	12/2013	Brandt et al.	N/A	N/A
2014/0243635	12/2013	Arefieg	N/A	N/A
2014/0335490	12/2013	Baarman et al.	N/A	N/A
2015/0120321	12/2014	David et al.	N/A	N/A
2015/0233749	12/2014	Wang et al.	N/A	N/A
2015/0342576	12/2014	Hall et al.	N/A	N/A
2015/0343173	12/2014	Tobescu et al.	N/A	N/A
2015/0359522	12/2014	Recht et al.	N/A	N/A
2015/0362351	12/2014	Joshi et al.	N/A	N/A
2016/0051176	12/2015	Ramos	600/573	G01F 23/265
2016/0183819	12/2015	Burnett et al.	N/A	N/A
2017/0035342	12/2016	Elia et al.	N/A	N/A
2017/0043089	12/2016	Handler	N/A	N/A
2017/0100068	12/2016	Kostov	N/A	N/A
2017/0113000	12/2016	Tobescu et al.	N/A	N/A
2017/0136209	12/2016	Burnett et al.	N/A	N/A
2017/0140103	12/2016	Angelides	N/A	A61F 5/4404
2017/0196478	12/2016	Hunter	N/A	N/A
2017/0202698	12/2016	Zani et al.	N/A	N/A
2017/0249445	12/2016	Devries et al.	N/A	N/A

2017/0290540	12/2016	Franco	N/A	N/A
2017/0291012	12/2016	Iglesias	N/A	N/A
2017/0307423	12/2016	Pahwa et al.	N/A	N/A
2017/0322197	12/2016	Hall et al.	N/A	N/A
2018/0015251	12/2017	Lampotang et al.	N/A	N/A
2018/0110456	12/2017	Cooper et al.	N/A	N/A
2018/0160961	12/2017	Gopinathan et al.	N/A	N/A
2018/0214122	12/2017	Ansell et al.	N/A	N/A
2018/0214297	12/2017	Hughett et al.	N/A	N/A
2018/0245967	12/2017	Parker et al.	N/A	N/A
2018/0280236	12/2017	Ludin et al.	N/A	N/A
2018/0317891	12/2017	Kim	N/A	N/A
2018/0344234	12/2017	Mckinney et al.	N/A	N/A
2019/0006047	12/2018	Gorek	N/A	G06F 18/25
2019/0017535	12/2018	Ormsbee et al.	N/A	N/A
2019/0046102	12/2018	Kushnir et al.	N/A	N/A
2019/0069829	12/2018	Bulut	N/A	N/A
2019/0069830	12/2018	Holt et al.	N/A	N/A
2019/0126006	12/2018	Rehm et al.	N/A	N/A
2019/0150821	12/2018	Waters et al.	N/A	N/A
2019/0167144	12/2018	Jung et al.	N/A	N/A
2019/0201596	12/2018	Luxon et al.	N/A	N/A
2019/0223844	12/2018	Aboagye et al.	N/A	N/A
2019/0247236	12/2018	Sides et al.	N/A	N/A
2019/0254582	12/2018	Wei et al.	N/A	N/A
2019/0321588	12/2018	Burnett et al.	N/A	N/A
2019/0328945	12/2018	Analytis et al.	N/A	N/A
2019/0343445	12/2018	Burnett et al.	N/A	N/A
2019/0358387	12/2018	Elbadry et al.	N/A	N/A
2019/0365308	12/2018	Laing et al.	N/A	N/A
2019/0381223	12/2018	Culbert et al.	N/A	N/A
2020/0022637	12/2019	Kurzrock et al.	N/A	N/A
2020/0064172	12/2019	Tabaczewski et al.	N/A	N/A
2020/0085378	12/2019	Burnett et al.	N/A	N/A
2020/0187863	12/2019	Tu et al.	N/A	N/A
2020/0268302	12/2019	Oh	N/A	N/A
2020/0268303	12/2019	Oliva	N/A	N/A
2020/0289749	12/2019	Odashima et al.	N/A	N/A
2020/0405524	12/2019	Gill	N/A	N/A
2021/0054610	12/2020	Hall et al.	N/A	N/A
2021/0077007	12/2020	Jouret et al.	N/A	N/A
2021/0100533	12/2020	Seres et al.	N/A	N/A
2021/0299353	12/2020	Mannu et al.	N/A	N/A
2022/0018692	12/2021	Tourchak et al.	N/A	N/A
2022/0026001	12/2021	Cheng et al.	N/A	N/A
2022/0026261	12/2021	Funnell et al.	N/A	N/A
2022/0079487	12/2021	Horiguchi et al.	N/A	N/A
2022/0192565	12/2021	Cheng et al.	N/A	N/A
2022/0192566	12/2021	Cheng et al.	N/A	N/A
2022/0193375	12/2021	Rehm et al.	N/A	N/A

2022/0233120	12/2021	Beuret et al.	N/A	N/A
2022/0296140	12/2021	Nguyen et al.	N/A	N/A
2022/0330867	12/2021	Conley et al.	N/A	N/A
2022/0386917	12/2021	Mann et al.	N/A	N/A
2023/0019703	12/2022	Behzad et al.	N/A	N/A
2023/0022547	12/2022	Cho et al.	N/A	N/A
2023/0025333	12/2022	Patel et al.	N/A	N/A
2023/0028966	12/2022	Franano	N/A	N/A
2023/0035669	12/2022	Raja et al.	N/A	N/A
2023/0040915	12/2022	Compton et al.	N/A	N/A
2023/0058553	12/2022	Fallows et al.	N/A	N/A
2023/0060232	12/2022	Patel et al.	N/A	N/A
2023/0084476	12/2022	Robichaud et al.	N/A	N/A
2023/0089041	12/2022	Handler	N/A	N/A
2024/0042120	12/2023	Cheng et al.	N/A	N/A
2024/0081708	12/2023	Kelly et al.	N/A	N/A
2024/0108268	12/2023	Woodard et al.	N/A	N/A
2024/0252783	12/2023	Waitkus et al.	N/A	N/A
2024/0347162	12/2023	Meese et al.	N/A	N/A
2024/0360938	12/2023	Cheng et al.	N/A	N/A
2024/0424186	12/2023	Justice et al.	N/A	N/A
2025/0090066	12/2024	Tourchak	N/A	N/A
2025/0120636	12/2024	Compton et al.	N/A	N/A
2025/0205456	12/2024	Rehm et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
2882654	12/2006	CA	N/A
2445749	12/2000	CN	N/A
200951235	12/2006	CN	N/A
201492414	12/2009	CN	N/A
102647939	12/2011	CN	N/A
103054559	12/2014	CN	N/A
107952140	12/2017	CN	N/A
109498013	12/2018	CN	N/A
110859636	12/2019	CN	N/A
112426156	12/2020	CN	N/A
0342028	12/1988	EP	N/A
2760470	12/2019	ES	N/A
2437549	12/2006	GB	N/A
2576743	12/2019	GB	N/A
S49-75171	12/1973	JP	N/A
S54-147066	12/1978	JP	N/A
S58-190719	12/1982	JP	N/A
S60-219517	12/1984	JP	N/A
H02-057240	12/1989	JP	N/A
H08-271301	12/1995	JP	N/A
H10-104041	12/1997	JP	N/A
2007-303982	12/2006	JP	N/A
2008-524618	12/2007	JP	N/A

2009-068959	12/2008	JP	N/A
2010-121950	12/2009	JP	N/A
2010-530978	12/2009	JP	N/A
2012-105947	12/2011	JP	N/A
2012-225790	12/2011	JP	N/A
2018108356	12/2017	JP	N/A
20070115495	12/2006	KR	N/A
2013740	12/2015	NL	N/A
2615727	12/2016	RU	N/A
1981003427	12/1980	WO	N/A
2004045410	12/2003	WO	N/A
2013013782	12/2012	WO	N/A
20130178742	12/2012	WO	N/A
2014/043650	12/2013	WO	N/A
2014105755	12/2013	WO	N/A
2014108690	12/2013	WO	N/A
2014/135856	12/2013	WO	N/A
2014/151068	12/2013	WO	N/A
2014145971	12/2013	WO	N/A
201511402	12/2014	WO	N/A
2015/105916	12/2014	WO	N/A
2015/127390	12/2014	WO	N/A
2015191125	12/2014	WO	N/A
2016177901	12/2015	WO	N/A
2017/023794	12/2016	WO	N/A
2018156624	12/2017	WO	N/A
2019066357	12/2018	WO	N/A
2019106675	12/2018	WO	N/A
2019/226697	12/2018	WO	N/A
2020033752	12/2019	WO	N/A
2020154370	12/2019	WO	N/A
2022108589	12/2021	WO	N/A
2022182794	12/2021	WO	N/A
2023022895	12/2022	WO	N/A
2023027871	12/2022	WO	N/A
2023076067	12/2022	WO	N/A

OTHER PUBLICATIONS

Bard Medical, Criticore Disposables—Non I.C., 3 pages, www.bardmedical.com/products/patienl-moniloring-, ystems/criticore®-system/criticore®-disposables-non-ic/ Jan. 30, 2015. cited by applicant
Bard Medical, Criticore Infection Control Disposables, 3 pages, www.bardmedical.com/patienl-moniloring-, ystems/criticore®-system/criticore®-infection-control-disposables/ Jan. 30, 2015. cited by applicant
Bard Medical, Criticore Monitor, 11 pages, www.bardmedical.com/products/patient-monitoring-systems/criticore®--monitor/ Jan. 30, 2015. cited by applicant
Bard Medical, Urine Meiers, 3 pages, www.bardmedical.com/products/urological-drainage/urine-collection/urinemeters/Jan. 30, 2015. cited by applicant
Biometrix, Urimetrix, 4 pages, www.biometrixmedical.com/Products/56/Urimetrix%E2%84%A2 Oct. 29, 2014. cited by applicant

Observe Medical, sippi, 3 pages, www.observemedical.com/products.html Oct. 29, 2014. cited by applicant
PCT/US19/33389 filed May 21, 2019 International Search Report and Written Opinion dated Aug. 2, 2019. cited by applicant
PCT/US2016/044835 filed Jul. 20, 2016 International Search Report and Written Opinion dated Dec. 16, 2016. cited by applicant
U.S. Appl. No. 17/054,493, filed Nov. 10, 2020 Final Office Action dated May 31, 2022. cited by applicant
U.S. Appl. No. 15/748,107, filed Jan. 26, 2018 Final Office Action dated Dec. 23, 2020. cited by applicant
U.S. Appl. No. 15/748,107, filed Jan. 26, 2018 Final Office Action dated Feb. 7, 2022. cited by applicant
U.S. Appl. No. 15/748,107, filed Jan. 26, 2018 Non-Final Office Action dated Sep. 3, 2021. cited by applicant
U.S. Appl. No. 15/748,107, filed Jan. 26, 2018 Non-Final Office Action dated Sep. 4, 2020. cited by applicant
U.S. Appl. No. 17/054,493, filed Nov. 10, 2020 Non-Final Office Action dated Nov. 24, 2021. cited by applicant
DFree Personal—Consumer Product Brochure, 2019. cited by applicant
DFree Pro Brochure 2019. cited by applicant
Leonhäuser, D et al., “Evaluation of electrical impedance tomography for determination of urinary bladder volume: comparison with standard ultrasound methods in healthy volunteers.”—BioMed Engr On-line; 17:95; 2018. cited by applicant
Li, R., et al., “Design of a Noninvasive Bladder Urinary vol. Monitoring System Based on Bio-Impedance.”—Engineering; vol. 5; pp. 321-325; 2013. cited by applicant
Reichmuth, M., et al., “A Non-invasive Wearable Bioimpedance System to Wirelessly Monitor Bladder Filling.”—Dep. of Health Sciences and Technology—Department of Information Technology and Electrical Engineering ETH Zurich, Zurich, Switzerland—Conference Paper; Mar. 2020. cited by applicant
SECA product catalog, https://US.secashop.com/products/seca-mbca/seca-mbca-514/5141321139, last accessed Sep. 11, 2020. cited by applicant
U.S. Appl. No. 17/054,493, filed Nov. 10, 2020 Final Office Action dated Oct. 4, 2023. cited by applicant
U.S. Appl. No. 17/262,080, filed Jan. 21, 2021 Final Office Action dated Sep. 11, 2023. cited by applicant
U.S. Appl. No. 17/262,080, filed Jan. 21, 2021 Notice of Allowance dated Oct. 13, 2023. cited by applicant
U.S. Appl. No. 17/306,821, filed May 3, 2021 Advisory Action dated Oct. 3, 2023. cited by applicant
U.S. Appl. No. 17/373,546, filed Jul. 12, 2021 Non-Final Office Action dated Nov. 1, 2023. cited by applicant
EP 20962628.2 filed May 31, 2023 Extended European Search Report dated Apr. 20, 2024. cited by applicant
EP 23188337.2 filed May 21, 2019 Extended European Search Report dated Dec. 4, 2023. cited by applicant
PCT/US2019/033389 filed Nov. 26, 2020 Extended European Search Report dated Jun. 4, 2021. cited by applicant
PCT/US2022/039191 filed Aug. 2, 2022 International Search Report and Written Opinion dated Dec. 5, 2022. cited by applicant
PCT/US2022/039746 filed Aug. 8, 2022 International Search Report and Written Opinion dated

Nov. 18, 2022. cited by applicant
PCT/US2022/046920 filed Oct. 17, 2022 International Search Report and Written Opinion dated Feb. 20, 2023. cited by applicant
U.S. Appl. No. 17/054,493, filed Nov. 10, 2020 Notice of Allowance dated Jan. 4, 2024. cited by applicant
U.S. Appl. No. 17/306,821, filed May 3, 2021 Notice of Allowance dated Apr. 23, 2024. cited by applicant
U.S. Appl. No. 17/373,546, filed Jul. 12, 2021 Notice of Allowance dated Mar. 7, 2024. cited by applicant
U.S. Appl. No. 17/373,546, filed Jul. 12, 2021 Notice of Allowance dated May 29, 2024. cited by applicant
U.S. Appl. No. 17/556,907, filed Dec. 20, 2021 Notice of Allowance dated Dec. 6, 2023. cited by applicant
U.S. Appl. No. 17/556,931, filed Dec. 20, 2021 Non-Final Office Action dated Mar. 27, 2024. cited by applicant
U.S. Appl. No. 17/556,931, filed Dec. 20, 2021 Restriction Requirement dated Feb. 22, 2024. cited by applicant
PCT/US2019/045787 filed Aug. 8, 2019 International Preliminary Report on Patentability dated Feb. 16, 2021. cited by applicant
PCT/US2019/045787 filed Aug. 8, 2019 International Search Report and Written Opinion dated Oct. 2, 2019. cited by applicant
PCT/US20/61367 filed Nov. 19, 2020 International Search Report and Written Opinion dated Feb. 22, 2021. cited by applicant
U.S. Appl. No. 17/262,080, filed Jan. 21, 2021 Non-Final Office Action dated Apr. 6, 2023. cited by applicant
U.S. Appl. No. 17/373,535, filed Jul. 12, 2021 Notice of Allowance dated Feb. 23. 2023. cited by applicant
U.S. Appl. No. 17/556,907, filed Dec. 20, 2021 Restriction Requirement dated May 12, 2023. cited by applicant
U.S. Appl. No. 15/748,107, filed Jan. 26, 2018 Notice of Allowance dated Dec. 12, 2022. cited by applicant
U.S. Appl. No. 17/054,493, filed Nov. 10, 2020 Non-Final Office Action dated Jan. 27, 2023. cited by applicant
U.S. Appl. No. 17/3026,821, filed May 3, 2021 Non-Final Office Action dated Jan. 10, 2023. cited by applicant
U.S. Appl. No. 17/373,535, filed Jul. 12, 2021 Non-Final Office Action dated Nov. 9. 2022. cited by applicant
“Urocare Reusable Night Drain Bottle—Urinary Collection System” Aug. 13, 2020, HealthProductsForYou.com, <https://www.healthproductsforyou.com/p-urocare-reusable-night-drain-bottle-urinary-collection-system.html> retrieved from Archive.org (Year: 2020). cited by applicant
U.S. Appl. No. 17/556,931, filed Dec. 20, 2021 Advisory Action dated Dec. 6, 2024. cited by applicant
U.S. Appl. No. 17/556,931, filed Dec. 20, 2021 Final Office Action dated Oct. 1, 2024. cited by applicant
U.S. Appl. No. 17/560,079, filed Dec. 22, 2021 Notice of Allowance dated Oct. 29. 2024. cited by applicant
PCT/US2022/017574 filed Feb. 23, 2022 Internation Search Report and Written Opinion dated Jun. 8, 2022. cited by applicant
Schlebusch, T. et al., “Bladder volume estimation from electrical impedance tomography”

Physiological Measurement Institute of Physics Publishing, Bristol, GB. vol. 35 No. 9 Aug. 20, 2014. (Aug. 20, 2014). cited by applicant
U.S. Appl. No. 17/306,821, filed May 3, 2021 Final Office Action dated Jul. 19, 2023. cited by applicant
U.S. Appl. No. 17/556,907, filed Dec. 20, 2021 Non-Final Office Action dated Aug. 17, 2023. cited by applicant
"Volumetric Flow Rate", www.vcalc.com/wiki/JeffNolumetric+%28Fluid%29+Flow+Rate, accessed Jan. 9, 2025, created Mar. 8, 2018 (Year: 2018). cited by applicant
U.S. Appl. No. 17/556,931, filed Dec. 20, 2021 Notice of Allowance dated Mar. 18, 2025. cited by applicant
U.S. Appl. No. 17/587,938, filed Jan. 28, 2022 Restriction Requirement dated Jan. 22, 2025. cited by applicant
U.S. Appl. No. 17/833,682, filed Jun. 6, 2022 Non-Final Office Action dated Jan. 15, 2025. cited by applicant
U.S. Appl. No. 17/870,698, filed Jul. 21, 2022 Restriction Requirement dated Feb. 12, 2025. cited by applicant
U.S. Appl. No. 17/879,658, filed Aug. 2, 2022 Non-Final Office Action dated Dec. 30, 2024. cited by applicant
U.S. Appl. No. 17/893,435, filed Aug. 23, 2022 Non-Final Office Action dated Jan. 17, 2025. cited by applicant
U.S. Appl. No. 17/587,938 filed Jan. 28, 2022 Non-Final Office Action dated May 12, 2025. cited by applicant
U.S. Appl. No. 17/682,785 filed Feb. 28, 2022 Restriction Requirement dated Apr. 2, 2025. cited by applicant
U.S. Appl. No. 17/833,682 filed Jun. 6, 2022 Final Office Action dated May 12, 2025. cited by applicant
U.S. Appl. No. 17/863,223 filed Jul. 12, 2022 Non-Final Office Action dated Apr. 2, 2025. cited by applicant
U.S. Appl. No. 17/870,698 filed Jul. 21, 2022 Non-Final Office Action dated Apr. 9, 2025. cited by applicant
U.S. Appl. No. 18/278,167 filed Aug. 21, 2023 Non-Final Office Action dated Apr. 24, 2025. cited by applicant
U.S. Appl. No. 17/863,923 filed Jul. 13, 2022 Restriction Requirement dated May 21, 2025. cited by applicant
U.S. Appl. No. 17/873,834 filed Jul. 26, 2022 Non-Final Office Action dated May 19, 2025. cited by applicant
U.S. Appl. No. 17/879,658 filed Aug. 2, 2022 Final Office Action dated May 14, 2025. cited by applicant
U.S. Appl. No. 17/883,507 filed Aug. 8, 2022 Restriction Requirement dated May 19, 2025. cited by applicant
U.S. Appl. No. 17/941,941 filed Sep. 9, 2022 Restriction Requirement dated May 28, 2025. cited by applicant
U.S. Appl. No. 18/036,335 filed May 10, 2023 Non-Final Office Action dated Jun. 18, 2025. cited by applicant
U.S. Appl. No. 18/682,075 filed Feb. 7, 2024 Non-Final Office Action dated Jun. 18, 2025. cited by applicant

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Background/Summary

PRIORITY (1) This application claims the benefit of priority to U.S. Provisional Application No. 63/127,041, filed Dec. 17, 2020, which is incorporated by reference in its entirety into this application.

SUMMARY
(1) Briefly summarized, embodiments disclosed herein are directed to systems, methods and apparatuses for automated measurement of a quantity of urine.
(2) One problem that often arises with collecting urine output (UO) data so clinicians can monitor and adjust the dosage of diuretics given to ambulatory patients is that conventional methods for collecting UO data (such as urine hats, bedside commodes, or bedpans) require patient compliance to urinate in the correct place. In many cases, patients are unable or unwilling to comply meticulously with their clinicians' instructions, or simply become fatigued or forget to comply as time goes on.
(3) A second problem is that conventional methods of collecting UO data require valuable clinical time to read and record the data accurately. For example, a nurse or other clinician may visually measure the amount of UO in a collection container like a urine hat, which consumes time and may be prone to human error. In addition, the clinician must record the data at regular intervals, for example every hour, in order to record a time series of UO measurements. This consumes additional valuable clinical time, while also increasing the chance of human error.
(4) A third problem is that indwelling Foley catheters, another option to accurately collect UO, can put patients at increased risk of catheter-associated urinary tract infections (CAUTIs). CAUTIs can lead to potentially dangerous complications, which can increase morbidity and even mortality in some cases, as well as causing discomfort and increasing treatment costs.
(5) Disclosed herein is a urine collection bag that can address these problems. The urine collection bag includes a collection area configured to collect urine received via tubing from a catheter. The urine collection bag further includes one or more force sensors coupled to or integrated into the urine collection bag. The one or more force sensors are configured to measure a pressure or a weight of the urine collected within the collection area. The urine collection bag further includes circuitry configured to determine the amount of the urine based at least on the pressure or weight measured by the one or more force sensors, and a specific gravity of the urine.
(6) In some embodiments, while determining the amount of the urine, the circuitry is further configured to determine that the pressure or weight measured by the one or more force sensors remains substantially constant for a predetermined time period. The circuitry is further configured to divide the pressure or weight by the specific gravity of the urine to determine an increase of a volume of the urine.
(7) In some embodiments, the circuitry receives the specific gravity of the urine as an input via a user interface.
(8) In some embodiments, the one or more force sensors comprise a resistive element operative to change resistance in response to a force. The resistance is measured by a printed circuit board (PCB).
(9) In some embodiments, the PCB further comprises a low-pass filter operative to filter noise.
(10) In some embodiments, the one or more force sensors comprise a plurality of force sensors.
(11) In some embodiments, the collection area comprises a flexible pouch. The one or more force

sensors measure the pressure of the urine by measuring an expansive force of the urine on one or more side walls of the flexible pouch.
(12) In some embodiments, the one or more force sensors are situated in an internal layer of the one or more side walls of the flexible pouch.
(13) In some embodiments, the catheter includes one or more of a small female external catheter (FEC) with an opening disposed on a top side, the opening configured to couple to a female anatomical part, a wicking catheter having a wicking area disposed on a top side, a finger-mountable catheter having a finger cavity configured to receive a user finger on a bottom side of the finger-mountable catheter, or a male external catheter (MEC).
(14) In some embodiments, the catheter is configured to remain on a patient while the patient stands or walks.
(15) In some embodiments, the urine collection bag further comprises a wireless transmitter configured to transmit the determined amount of the urine to an electronic medical records system.
(16) In some embodiments, the circuitry comprises a printed circuit board (PCB) or a processor.
(17) In some embodiments, a respective force sensor of the one or more force sensors is powered by a battery.
(18) Also disclosed herein is a urine collection system. The urine collection system comprises a catheter, a urine collection bag configured to measure an amount of urine, and flexible drainage tubing coupling the catheter to the urine collection bag. The urine collection bag comprises a collection area configured to collect the urine, wherein the urine is received via the flexible drainage tubing from the catheter. The urine collection bag further comprises one or more force sensors coupled to or integrated into the urine collection bag. The one or more force sensors are configured to measure a pressure or a weight of the urine collected within the collection area. The urine collection bag further comprises circuitry configured to determine the amount of the urine based at least on the pressure or weight measured by the one or more force sensors, and a specific gravity of the urine.
(19) In some embodiments, the flexible drainage tubing is secured to a leg of a patient with a stabilization device comprising an adhesive pad, the adhesive pad comprising one or more adhesive wings, and a retainer configured to stabilize the flexible drainage tubing. Alternatively, the flexible drainage tubing is secured to a leg of a patient with a fabric strap wrapped around the leg.
(20) In some embodiments, the urine collection system further comprises a wireless transmitter configured to transmit the determined amount of the urine to an electronic medical records system.
(21) Also disclosed herein is a method of using a urine collection bag and a catheter to measure an amount of urine. The method comprises receiving the urine, by the urine collection bag, via tubing from the catheter. The method further comprises collecting the urine, by the urine collection bag, in a collection area of the urine collection bag. The method further comprises measuring, by one or more force sensors coupled to or integrated into the urine collection bag, a pressure or a weight of the urine collected within the collection area. The method further comprises determining the amount of the urine, by circuitry coupled to or integrated into the urine collection bag, and based at least on the pressure or weight measured by the one or more force sensors, and a specific gravity of the urine.
(22) In some embodiments, the tubing connecting the catheter to the urine collection bag is secured to a leg of the patient with a stabilization device. The stabilization device comprises an adhesive pad comprising one or more adhesive wings and a retainer configured to stabilize the flexible drainage tubing. Alternatively, the tubing connecting the catheter to the urine collection bag is secured to a leg of the patient with a fabric strap wrapped around the leg.
(23) In some embodiments, the catheter is configured to remain on the patient while the patient stands or walks.
(24) These and other features of the concepts provided herein will become more apparent to those

of skill in the art in view of the accompanying drawings and following description, which disclose particular embodiments of such concepts in greater detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the disclosure are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings, in which like references indicate similar elements and in which:

(2) FIG. **1** illustrates a urine collection system, according to some embodiments;

(3) FIG. **2**A displays details of a urine collection bag, according to some embodiments;

(4) FIG. **2**B displays the functioning of a force sensor in a urine collection bag, according to some embodiments;

(5) FIG. **2**C displays a configuration of force sensors in a layer within the walls of a urine collection bag, according to some embodiments;

(6) FIG. **3**A illustrates details of a female external catheter (FEC), according to some embodiments;

(7) FIG. **3**B illustrates details of a wicking catheter, according to some embodiments;

(8) FIG. **3**C illustrates details of a finger-mountable catheter, according to some embodiments;

(9) FIG. **3**D illustrates details of a male external catheter (MEC), according to some embodiments;

(10) FIG. **4**A illustrates a urine collection bag strapped to a patient's leg via a fabric strap, according to some embodiments;

(11) FIG. **4**B illustrates a urine collection bag secured to a patient's leg via a stabilization device, according to some embodiments; and

(12) FIG. **5** shows a flowchart of an example method for measuring an amount of urine with a urine collection bag, according to some embodiments.

DETAILED DESCRIPTION

(13) Before some particular embodiments are disclosed in greater detail, it should be understood that the particular embodiments disclosed herein do not limit the scope of the concepts provided herein. It should also be understood that a particular embodiment disclosed herein can have features that can be readily separated from the particular embodiment and optionally combined with or substituted for features of any of a number of other embodiments disclosed herein.

(14) Regarding terms used herein, it should also be understood the terms are for the purpose of describing some particular embodiments, and the terms do not limit the scope of the concepts provided herein. Ordinal numbers (e.g., first, second, third, etc.) are generally used to distinguish or identify different features or steps in a group of features or steps, and do not supply a serial or numerical limitation. For example, “first,” “second,” and “third” features or steps need not necessarily appear in that order, and the particular embodiments including such features or steps need not necessarily be limited to the three features or steps. Labels such as “left,” “right,” “top,” “bottom,” “front,” “back,” and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to reflect, for example, relative location, orientation, or directions. Singular forms of “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(15) With respect to “proximal,” a “proximal portion” or a “proximal end portion” of, for example, a probe disclosed herein includes a portion of the probe intended to be near a clinician when the probe is used on a patient. Likewise, a “proximal length” of, for example, the probe includes a length of the probe intended to be near the clinician when the probe is used on the patient. A “proximal end” of, for example, the probe includes an end of the probe intended to be near the clinician when the probe is used on the patient. The proximal portion, the proximal end portion, or the proximal length of the probe can include the proximal end of the probe; however, the proximal

portion, the proximal end portion, or the proximal length of the probe need not include the proximal end of the probe. That is, unless context suggests otherwise, the proximal portion, the proximal end portion, or the proximal length of the probe is not a terminal portion or terminal length of the probe.

(16) With respect to "distal," a "distal portion" or a "distal end portion" of, for example, a probe disclosed herein includes a portion of the probe intended to be near or in a patient when the probe is used on the patient. Likewise, a "distal length" of, for example, the probe includes a length of the probe intended to be near or in the patient when the probe is used on the patient. A "distal end" of, for example, the probe includes an end of the probe intended to be near or in the patient when the probe is used on the patient. The distal portion, the distal end portion, or the distal length of the probe can include the distal end of the probe; however, the distal portion, the distal end portion, or the distal length of the probe need not include the distal end of the probe. That is, unless context suggests otherwise, the distal portion, the distal end portion, or the distal length of the probe is not a terminal portion or terminal length of the probe.

(17) The term "logic" may be representative of hardware, firmware or software that is configured to perform one or more functions. As hardware, the term logic may refer to or include circuitry having data processing and/or storage functionality. Examples of such circuitry may include, but are not limited or restricted to a hardware processor (e.g., microprocessor, one or more processor cores, a digital signal processor, a programmable gate array, a microcontroller, an application specific integrated circuit "ASIC", etc.), a semiconductor memory, or combinatorial elements.

(18) Additionally, or in the alternative, the term logic may refer to or include software such as one or more processes, one or more instances, Application Programming Interface(s) (API), subroutine(s), function(s), applet(s), servlet(s), routine(s), source code, object code, shared library/dynamic link library (dll), or even one or more instructions. This software may be stored in any type of a suitable non-transitory storage medium, or transitory storage medium (e.g., electrical, optical, acoustical or other form of propagated signals such as carrier waves, infrared signals, or digital signals). Examples of a non-transitory storage medium may include, but are not limited or restricted to a programmable circuit; non-persistent storage such as volatile memory (e.g., any type of random access memory "RAM"); or persistent storage such as non-volatile memory (e.g., read-only memory "ROM", power-backed RAM, flash memory, phase-change memory, etc.), a solid-state drive, hard disk drive, an optical disc drive, or a portable memory device. As firmware, the logic may be stored in persistent storage.

(19) Clinicians often require accurate urine output (UO) data in order to monitor and adjust the dosage of diuretics given to ambulatory patients. Conventional methods for collecting UO data include urine hats, bedside commodes, and bedpans. However, these methods require patient compliance to urinate in the correct place. These methods also consume valuable clinician time to accurately read and record the data. Another existing option to accurately collect UO is indwelling Foley catheters. However, such indwelling catheters can put patients at increased risk of catheter-associated urinary tract infections (CAUTIs).

(20) The disclosed urine collection system and methods can address these problems by automating accurate measurement of UO while a patient sits, stands, walks, and goes about other activities. As a result, the disclosed system and methods can measure UO more continually and accurately than conventional systems, while also being less disruptive to patients' lives.

(21) Referring to FIG. **1**, a urine collection system **100** is illustrated, according to some embodiments. System **100** includes a catheter **110**, a urine collection bag **120** that can measure the weight and/or volume of urine it contains, and tubing **130** coupling catheter **110** to urine collection bag **120**. The urine collection bag **120** comprises a collection area configured to collect the urine received via tubing **130**. In an embodiment, the urine collection bag **120** may be inside a rigid container supporting the bottom and/or side walls of collection bag **120**. Alternatively, the urine collection bag **120** may attach to a wheelchair, a bed, or a patient's leg.

(22) The urine collection bag **120** further comprises a network of force sensors **140**, which are coupled to or integrated into collection bag **120**. The force sensors **140** can measure forces, pressure, or weight of the urine in the collection area. The urine collection bag **120** can also include circuitry that can compute the amount of urine based on the pressure or weight measured by the network of force sensors **140**, taking into account information about the urine's specific gravity. For example, system **100** can determine the volume of urine contained in urine collection bag **120** by dividing forces associated with the urine's weight or pressure by the urine's specific gravity, as disclosed herein below. The circuitry may include a small printed circuit board (PCB) and/or a processor, which may be powered by a battery. The urine collection bag **120** and sensors **140** will be described further in the examples of FIGS. **2**A-**2**C, catheter **110** will be described further in the examples of FIGS. **3**A-**3**D, and details relevant to tubing **130** will be described further in the examples of FIGS. **4**A-**4**B below.
(23) In addition, system **100** can include a wireless transmission technology, such as BLUETOOTH® and/or Wi-Fi technology, to transmit the patient's UO data to an electronic medical records (EMR) system of a hospital or other facility. Of course, a wired connection may also be used in some embodiments.
(24) The urine collection system **100** has a number of advantages over conventional methods and systems for measuring UO. In a typical embodiment, catheter **110** is external, so it poses lower risk of causing a catheter-associated urinary tract infection (CAUTI) than conventional indwelling Foley catheters. Further, as the urine collection bag **120** automatically measures the volume of urine, it eliminates human error that may occur when a nurse or other clinician visually measures the amount of UO in a collection container like a urine hat. System **100** can automatically transmit the UO data to an electronic medical records (EMR) system of a hospital or other facility, thereby saving time for clinicians because they do not need to measure patients' UO every hour. A clinician only needs to place catheter **110** once, and it can remain on the patient for the entire duration of the patient's stay. Moreover, system **100** virtually ensures patient compliance, by contrast with conventional systems, which require patients to urinate in a specific receptacle (such as a conventional urine hat or commode).
(25) FIG. **2**A displays the assembly **200** of the urine collection bag **120**, according to some embodiments. In this example, the collection bag **120** contains a patient's urine, which can be measured by pressure sensor **210** on side wall **215** of collection bag **120**, and/or by weight sensor **220**, which may be located on a bottom or underside **225** of the collection bag **120**. Sensors **210** and **220** can be coupled to or integrated into the urine collection bag **120**, and are used to determine the amount of urine in bag **120**. By automatically measuring the volume of urine, urine collection bag **120** improves accuracy and eliminates human error that may occur when a clinician visually measures the amount of UO in a conventional collection container, such as a urine hat.
(26) In some embodiments, the urine collection bag **120** can include a network of multiple sensors located on the side **215** of and/or the underside **225** of the collection bag **120**. In some embodiments, these multiple sensors can produce multiple force and/or pressure measurements that can be combined, either by simple averaging or by a more sophisticated algorithm or machine learning (ML) method. In some embodiments, the multiple sensors may operate synergistically such that collective information may be determined from the measurements/readings from a plurality of sensors that would otherwise be incomplete or unobtainable from just a single sensor. For example, the system **100** may determine which sensor readings are most reliable at any given time, e.g., as the collection bag **120** tilts, moves, and/or deforms during usage, and weight the individual sensor readings accordingly. In one example, such movement may be detected by a plurality of sensors **220** located on the underside **225** of the urine collection bag **120** when one or more sensors **220** measure a weight that differs from other sensors **220**). In a second example, if the collection bag **120** tilts in a way such that sensor **210** no longer experiences pressure from the urine, or sensor **220** no longer experiences the weight of the urine, the system may temporarily

reduce or eliminate its reliance on readings from those sensors.

(27) In various embodiments, force sensors **210** and **220** may include one or more strain gauge sensors, pressure sensors, weight sensors, resistive elements, and the like, and are not limited by the present disclosure. All these specific forms of sensors may be referred to herein interchangeably as force sensors, or simply as sensors. In an embodiment, pressure sensor **210** may measure a pressure and/or strain exerted by side wall **215** due to the urine inside bag **120**, as described in the example of FIG. **2**B below. In an embodiment, weight sensor **220** directly measures the weight of urine in bag **120**.

(28) In some embodiments, little conversion is necessary from the pressure, weight, force, electrical resistance, and the like measured by sensors **210** or **220**. For example, in the case of weight sensor **220**, the sensor may measure the weight of the urine, which is a direct way to measure the quantity of UO. That is, the system may treat the force measurement of weight sensor **220** as a direct measurement of the amount of urine, for example by reporting the weight measurement of the urine (or the equivalent mass of the urine) directly to an EMR system of a hospital or other facility.

(29) Alternatively, in some embodiments, conversion is necessary to determine the quantity of UO from the force sensor measurements. Accordingly, the system may include circuitry, such as a PCB and/or a processor powered by a battery, that can convert the measurements to a volume of urine based on the urine's specific gravity. For example, in the case of weight sensor **220**, the circuitry may divide the weight, or the equivalent mass, by the urine's specific gravity or density to compute a volume. Likewise, in the case of pressure or strain sensor **210**, the circuitry may need to compute the UO based on a measured pressure or strain on side wall **215** of collection bag **120**. This may also be accomplished by dividing the measured force by the urine's specific gravity or density, as will be described further in the example of FIG. **2**B below. Such conversion may be performed by hardware, firmware, and/or software implemented by the circuitry, PCB, and/or processors of the urine collection system, and is not limited by the present disclosure.

(30) The specific gravity of the urine may be approximately 1.020, such as between 1.000 and 1.050, or more specifically between 1.010 and 1.030. In an embodiment, the system may receive the specific gravity as an input, for example from a nurse. The nurse may be able to change the specific gravity when needed, for example for use with a different patient, or when a patient's diet or health changes. In various embodiments, the system may receive the specific gravity via a user interface, or via Bluetooth or Wi-Fi technology from a local device, a local network, or an Internet cloud server. Patient-specific information, such as the specific gravity of the urine, may then be stored in transitory memory or non-transitory storage associated with urine collection bag **120** and/or the urine collection system, and is not limited by the present disclosure.

(31) It should be noted that in some embodiments, the specific gravity of the collected urine may be determined based on at least the sensor measurements. In such embodiments, the circuitry of the system may determine and specific gravity of the collected urine and provide an alert or indication when the specific gravity is outside of a predetermined range. For example, a specific gravity outside of the range of 1.010 to 1.030 may indicate the patient's acute condition (e.g., dehydration) or a disease process progression (e.g., kidney failure).

(32) FIG. **2**B displays the functioning of a force sensor **210** located on a side wall **215** of urine collection bag **120**, according to some embodiments. In this example, urine collection bag **120** is filled with urine **250**, which exerts a pressure on side wall **215** of bag **120**. Side wall **215** may expand and/or strain against force sensor **210**. For example, in some embodiments, bag **120** is made of vinyl or another flexible material, so that side wall **215** can push against force sensor **210**. In another example, bag **120** comprises a pouch, and as the bag fills, side wall **215** stretches or has an increased amount of pressure applied thereto.

(33) In some embodiments, a countervailing force may be present on force sensor **210**. Such a countervailing force may be necessary to stabilize force sensor **210**, so that it does not accelerate or

move, and can therefore measure the strain or pressure accurately. This force may be provided by various sources in various embodiments, for example from a stabilizing structure or element such as a fabric or elastic strap or reusable collection bag cover wrapped around the collection bag (see FIGS. **4**A and **4**B). Similarly, the countervailing force may be provided by an external surface, such as the patient's leg (see FIGS. **4**A and **4**B), another body part, or furniture such as a chair. In another example, the side walls **215** of collection bag **120** may themselves provide countervailing force to the pressure or strain from the urine, for example if side walls **215** are massive or stiff enough. In a related example, the force sensors may be contained within an internal layer of side walls **215** and/or collection bag **120** (see FIG. **2**C). In the latter case, the exterior layer of side walls **215** and/or collection bag **120** may provide the countervailing force. In yet another example, bag **120** may be inside a rigid container that supports the bottom and walls of the bag, and moreover provides a countervailing force against the pressure of urine **250**. Alternatively, the urine collection bag **120** may attach to a wheelchair, bed, or a patient's leg, which may also provide countervailing force. Accordingly, force sensor **210** can be stabilized so as to measure the strain or pressure exerted by the urine **250**. The system then uses this measurement to determine the UO, as described herein below.

(34) In various embodiments, the system's sensors, such as sensor **210**, may include one or more strain gauge sensors, pressure sensors, weight sensors, resistive elements, and the like, and are not limited by the present disclosure. All these specific forms of sensors may be referred to herein interchangeably as force sensors. Force sensor **210** may comprise resistive elements with electrical resistance that changes in response to force. In particular, the force sensors may be strain gauge sensors, for example comprising piezoresistors or metallic foil elements that deform in response to strain, thereby changing the electrical resistance of the sensors. The change in the electrical resistance of sensor **210** may be measured by a printed circuit board (PCB), for example using a Wheatstone bridge, thereby measuring the force on sensor **210**. In some embodiments, sensor **210** and the PCB may be powered by a battery and/or a rechargeable battery, so that the urine collection system is portable and convenient. The force sensors may be precise, for example they may be able to measure to within ±2 grams or ±0.02 Newtons, the accuracy needed for treating critical patients. Example force sensors include Model 1075 by Adafruit, Sensor-Puck by Silicon Labs, Model FSR06BE by Ohmite, and FSR 400 Series Force-Sensing Resistors by Interlink Electronics.

(35) In some embodiments, the PCB and/or circuitry can also include Bluetooth and/or Wi-Fi technology to transmit the patient's UO data to an electronic medical records (EMR) system of a hospital or other facility. Such a feature can save time for clinicians because they do not need to measure patients' UO every hour.

(36) In some embodiments, the system applies Bernoulli's equation to determine the variation of fluid pressure with depth, and/or to convert the force measurements to a volume of UO. For example, force measurement logic implemented by the circuitry, PCB, or processor can apply Bernoulli's equation to perform the conversion. Fluids such as water or urine, which has a similar specific gravity to water, may be approximately incompressible at typical room temperatures and ambient air pressures. Therefore, the system may apply a simplified form of Bernoulli's equation for an approximately incompressible fluid in equilibrium, i.e., with no net flow. In particular, Bernoulli's equation states that the pressure $P(z)$ varies as a function of height z in a column of incompressible fluid of density ρ because each layer of fluid must support all the fluid above it: $P(z)=P(z=0)-\rho g z$, where $P(z=0)$ is the pressure at the bottom of the column, and $g=9.8\ m/s^2$ is the acceleration due to gravity. Note that the fluid's specific gravity is ρ divided by the density ρ_{H2O} of water. Defining $P(z=z_{max})$ as the pressure at the top z_{max} of the column of fluid, and $d=z_{max}-z$ as the depth within the column, this is equivalent to $P(z)=P(z=z_{max})+\rho g d$.

(37) Accordingly, if a pressure sensor **210** is located at a specific height z_{sens} on the wall **215** of urine collection bag **120**, it will experience an increase in pressure proportional to the amount ρ

d of fluid above z.sub.sens. Specifically, setting z.sub.max as the top of the column of fluid, then P(z=z.sub.max)=P.sub.atm is atmospheric pressure, and the depth d=z.sub.max−z.sub.sens is a measure of the amount of fluid above z.sub.sens. Accordingly, the pressure P at depth d, the location of sensor **210**, is increased over P.sub.atm by an amount ρ g d proportional to the amount of fluid higher than sensor **210**.
(38) Note that this equation may remain correct even if side walls **215** are not vertical. Side walls **215** may not be vertical, both because collection bag **120** may itself have, for example, a convex shape, and because the pressure of the urine **250** may deform walls **215**, particularly if the walls are made from a flexible material like vinyl. Thus, even though an incompressible fluid's density is constant, the lateral extent of the fluid at each height z may depend on z. If z.sub.max is located in a portion of the bag in which the bag's breadth increases with z, the fluid's top surface will still be unconstrained by the bag, and the above equations still hold for z.sub.sens below z.sub.max. If z.sub.max is located in a portion of the bag in which the bag's breadth decreases with z, the depth d of fluid above z.sub.sens will vary with the lateral coordinates x and y, because at some locations the fluid's top surface is constrained by the bag walls. In equilibrium, the pressure P will still equilibrate at each height z in the fluid, and the bag walls will provide a force equivalent to the weight of the missing fluid. Accordingly, the pressure P(z) is still P(z.sub.max)+ρ g d, where d is now measured from z to the highest point z.sub.max of the fluid at any lateral location (x, y). However, in the case of non-vertical walls, computing the volume of UO will require a correction for the expansion of the bag. In some embodiments, the system can model the bag's shape numerically. For example, the system may use interpolation based on empirical measurements of the bag's shape as a function of the volume of fluid.
(39) In some embodiments, the force sensor does not require continuous calibration, because many commercially-available sensors, such as those mentioned above, are pre-calibrated. Force measurement logic implemented by the circuitry, PCB, or processor can be instructed to convert the measured pressure, force, or weight to volume of UO. In some embodiments, a single urine collection bag may be used for multiple patients in succession. In this case, it is preferable to perform a calibration check before using the urine collection bag with a new patient, so as to ensure the accuracy of the force sensor, and to ensure that the force sensor is still properly calibrated.
(40) There are several ways the system can reduce or eliminate unwanted noise, such as spurious or stray forces on the bag, and prevent the noise from being recorded erroneously as UO. In some embodiments, a delay period can be implemented by the force measurement logic, so as to require at least a threshold period of quiet time before it records a force reading and converts this to a UO volume measurement. For example, suppose the force sensor senses a possibly transient mechanical impulse from the urine collecting into the bag **120**. In fact, such a measured impulse could be due to new urine arriving in the bag **120**, or to sloshing motion of the existing urine, or to any other transient noise source, and it may be difficult for the sensor to immediately differentiate among these scenarios. Accordingly, the force sensors and/or circuitry, PCB, or a processor associated with the urine collection bag **120** can be instructed to wait for the sensed force to stabilize, and remain substantially constant for a threshold period (e.g., 5, 10, or 15 seconds), before recognizing the force as a bonafide non-transient signal.
(41) In some embodiments, noise can be filtered electronically so that only UO measurements are recorded. For example, the PCB can include a low pass RC filter to filter noise, which may originate in stray forces or in electronic noise within the circuit. For example, a low-pass RC filter with a cutoff frequency of 650 Hz may be used, such as a 750 ohm resistor and a 0.33 g capacitor. This filter may use a DC source impedance, which is generally compatible with a successive approximation analog-to-digital (A/D) converter of a microcontroller. This filter may be integrated into the PCB that controls the force sensors.
(42) FIG. **2**C displays a configuration of force sensors in a layer within the walls **270** of a urine collection bag, according to some embodiments. In this example, the force sensors can be

integrated into a middle layer **275**, sandwiched between exterior layer **280** and interior layer **285** of the urine collection bag. Interior layer **285** and exterior layer **280** can comprise a strong, flexible, watertight material, such as vinyl. Middle layer **275** may include all the instruments and/or electronics needed to measure the pressure or weight of the urine in the bag and to convert these measurements to the volume of UO. For example, middle layer **275** can include force sensor **210** of the example of FIG. **2**B, circuitry, a PCB, and/or a processor. In an embodiment, the middle layer **275** can also include a battery to power the force sensors, circuitry, PCB, and/or processor. By containing all these elements in middle layer **275**, the sandwich structure of wall **270** can protect sensitive instrumentation and/or electronics from the patient's urine, sweat, dirt, and any other contaminants. Moreover, the additional exterior side wall layer **280** can provide a countervailing force that stabilizes the force sensors in middle layer **275**, making it possible to measure the pressure of the urine accurately. In an alternative embodiment, as described in the example of FIG. **2**B, the force sensors may be outside the urine collection bag, and are not limited by the present disclosure.

(43) The external catheter is also an important part of the urine collection system. Referring to FIG. **3**A, details of a female external catheter (FEC) **300** are illustrated, according to some embodiments. FEC **300** includes coupled vessels **305** and **325**, and can be connected via Luer connector **310** to drainage tubing **130**, which leads to the urine collection bag. As shown, vessel **325** is curved upward to opening **330**, which can receive urine from the patient's urethra. In an embodiment, the shape of the FEC **300** can be focused on coupling to the patient's urethra, and FEC **300** can be small and be held in place well due to its proximity to the patient anatomy.

(44) In an embodiment, FEC **300** may be held in place by the female anatomy (for example, by the labia or an extension into the vaginal opening). In particular, the shape of FEC **300** can be designed to couple well enough to the patient anatomy, such that FEC **300** can remain in place and operational while the user comfortably lies down, sits (e.g., in a wheelchair, chair, or bed), stands, and even walks. Accordingly, a clinician only needs to place FEC **300**, wicking catheter **340** (see FIG. **3**B), finger-mountable catheter **360** (see FIG. **3**C), or male external catheter **380** (see FIG. **3**D) once, and it can remain on the patient for the entire duration of the patient's stay.

(45) Because such external catheters are comfortable and can remain in place, the disclosed urine collection system virtually ensures patient compliance, by contrast with conventional systems, which require patients to urinate in a specific receptacle (such as a conventional urine hat or commode). Moreover, because FEC **300**, wicking catheter **340**, and finger-mountable catheter **360** are external, they pose lower risk of causing a CAUTI than an indwelling Foley catheter.

(46) FIG. **3**B illustrates details of a wicking catheter **340**, according to some embodiments. Wicking catheter **340** has a wicking area **345**, which may be comprised of gauze. In an example, wicking catheter **340** may be the PureWick (TM) catheter distributed by Bard Medical Devices. In some embodiments, wicking catheter **340** can be used together with a low pressure wall suction pump to wick urine gently away from the patient via wicking area **345**, and into the urine collection bag. Alternatively, wicking catheter **340** may be an intermittent catheter, such as the intermittent catheters described in U.S. Provisional Application No. 63/060,615, filed Aug. 3, 2020, and titled "Intermittent-Catheter Assembly and Methods Thereof," and U.S. Provisional Application No. 63/077,469, filed Sep. 11, 2020, and titled "Intermittent-Catheter Assembly and Methods Thereof," each of which is incorporated by reference in its entirety into this application.

(47) FIG. **3**C illustrates details of a finger-mountable catheter **360**, according to some embodiments. In one example, finger-mountable catheter **360** also has a wicking area **365**, which may be comprised of gauze. In addition, finger-mountable catheter **360** has a finger pocket **370** located on a bottom side of finger-mountable catheter **360**. The finger pocket **370** may be a cavity that is configured to receive a user's finger. Once the user has inserted her finger into the finger pocket **370**, she may manipulate the placement and orientation of finger-mountable catheter **360** such that wicking area **365** is placed over the patient's urethra in order to collect or "wick" urine,

which is passed to the catheter tubing **130**, and from there to the urine collection bag. Finger-mountable catheters are discussed further in U.S. Provisional Application No. 63/115,564, filed Nov. 18, 2020, and titled "External Catheter with Finger Pocket for Self-Placement," which is incorporated by reference in its entirety into this application.

(48) FIG. **3**D illustrates details of a male external catheter (MEC) **380**, according to some embodiments. While the catheters of the examples of FIGS. **3**A-**3**C above are female-only solutions, the disclosed system and methods can also be applied to males, for example using MEC **380**, such as a so-called "condom-catheter." MEC **380** can also remain in place and operational while the user lies down, sits (e.g., in a wheelchair, chair, or bed), stands, and walks. In some embodiments, MEC **380** may include features to prevent leakage of urine, for example by improving directionality of flow or ameliorating gaps in the flow passage, thereby improving the accuracy of UO measured by the collection bag. Because MEC **380** is external, it poses a lower risk of causing a CAUTI than an indwelling Foley catheter.

(49) The disclosed system addresses the need for clinicians to collect UO data, for the purpose of monitoring and adjusting the dosage of diuretics given to ambulatory patients, more accurately and less invasively than conventional systems. As described in the examples of FIGS. **3**A-**3**D, the catheter can be designed to remain on a patient while the patient stands, walks, and goes about other activities. Accordingly, the disclosed system and methods can make accurate measurements of UO on a more continual basis than conventional systems, all while providing less disruption to patients' lives than conventional systems.

(50) In some embodiments, the disclosed urine collection system can include means to secure and stabilize the urine collection bag. In particular, a reusable collection bag cover, fabric strap, or stabilization device that stabilizes the urine collection bag can be an important part of enabling the patient to go about normal activities, while fulfilling the need to monitor UO with very high accuracy.

(51) Referring to FIG. **4**A, a urine collection bag **120** strapped to a patient's leg via a fabric strap **400** is illustrated, according to some embodiments. In this example, the urine collection bag is placed on the patient's leg and is secured in place with fabric strap **400**, which is wrapped around the patient's leg. Strap **400** can enable the patient to walk while wearing the catheter and collection bag, and while using the system to measure UO. Fabric strap **400** may also function as a reusable cover for the collection bag.

(52) In an embodiment, the placement of the collection bag is not crucial to the system's operation, since the urine is collected from the patient by the catheter, and is then passed via the flexible drainage tubing **130** to the collection bag. Accordingly, key functions of strap **400** can be to secure the collection bag in proximity to the catheter, to conceal the urine in the collection bag, and to prevent the collection bag from shifting, spilling, being shaken, etc. However, in some embodiments, proper measurement of the UO may depend at least somewhat on satisfactory orientation or positioning of the collection bag. For example, it may be necessary to ensure the urine collection bag is not upside-down in order for the force sensors to measure force properly, particularly in the case where a sensor on the underside **225** of the collection bag measures the weight of the urine, such as sensor **225** in the example of FIG. **2**A. In this case, the collection bag can be placed with the proper position and/or orientation, and secured with fabric strap **400**.

(53) In some embodiments, the fabric strap **400** may also contribute to the proper functioning of the collection bag, and in particular the force sensors. For example, as described in the example of FIG. **2**B above, when urine flows into the collection bag, it can expand the collection bag so as to press against the force sensors. In an embodiment, the force sensors, or a PCB containing force sensors, can be located between the collection bag and fabric strap **400**, which may be a reusable collection bag cover. Accordingly, fabric strap **400** may provide a countervailing force against the pressure from the urine in the collection bag. This countervailing force, in turn, may stabilize the force sensors and help obtain a more reliable force measurement. In an embodiment, the force sensors or

PCB can be integrated into fabric strap **400**. In an embodiment, the collection bag may be designed with layers, similar to the example of FIG. **2**C, but fabric strap **400** may form an outer layer, rather than the outer vinyl layer **280** of FIG. **2**C.

(54) Tubing **130** may be flexible and durable enough to enable the patient to stand, walk, and perform normal activities comfortably, without damaging tubing **130**. In some embodiments, tubing **130** may comprise plastic. For example, tubing **130** may be the same tubing used for conventional urine bags that are placed on a patient's leg.

(55) FIG. **4**B illustrates flexible drainage tubing **130** of a urine collection system secured to a patient's leg via a stabilization device **450**, according to some embodiments. In some embodiments, the flexible drainage tubing **130** is secured to the patient's leg with a stabilization device **450** comprising an adhesive pad and a retainer configured to stabilize the flexible drainage tubing **130**. The adhesive pad can include one or more adhesive wings, for example two adhesive wings **460** as shown. In various embodiments, stabilization device **450** may be adhered and/or sutured to the patient's skin. Additional medical tape **470** may be used to secure the tubing **130**. In an example, stabilization device **450** may be the STATLOCK® stabilization device distributed by C.R. Bard Inc. Stabilization device **450** has advantages over conventional stabilization methods, such as improved stability and retention, anatomical conformity, and adhesive strength. This can help ensure that the tubing **130** remains orderly and close to the patient, and reduces the risk of kinks, tangling, or breakage of tubing **130**. In some embodiments, both fabric strap **400** and stabilization device **450** can be used together to stabilize both the collection bag and tubing **130**.

(56) Referring to FIG. **5**, a flowchart is shown of an example method **500** for measuring an amount of urine with a urine collection bag, according to some embodiments. Each block illustrated in FIG. **5** represents an operation performed in the method **500** of measuring an amount of urine with a urine collection bag. The method can be performed by a urine collection system, such as urine collection system **100** in the example of FIG. **1** above.

(57) As an initial step in the method **500**, the system can optionally receive the specific gravity as an input, for example from a clinician (block **510**). The clinician may be able to input the specific gravity whenever needed, for example for use with a different patient, or when a patient's diet or health changes. In various embodiments, the system may receive the specific gravity via a user interface, or via a Bluetooth or Wi-Fi transmission from a local device, a local network, or an Internet cloud server. The system can subsequently use the specific gravity to compute the volume of UO based on force measurements. The specific gravity of the urine may be similar to that of water. For example, the specific gravity may be approximately 1.020, such as between 1.000 and 1.050, or more specifically between 1.010 and 1.030.

(58) Next, the system can detect that a force is present (block **520**). For example, the force sensors can detect a change in the pressure or weight of urine in the urine collection bag, as described in the example of FIG. **2**B above. In some embodiments, the system can filter noise electronically so that only forces due to urine output are recorded. For example, a low-pass RC filter with a cutoff frequency of 650 Hz may be used, such as a 750 ohm resistor and a 0.33 μF capacitor. This filter may use a DC source impedance, which is generally compatible with a successive approximation analog-to-digital (A/D) converter of a microcontroller. This filter can be integrated into the PCB that controls the force sensors.

(59) Next, the system can determine that the force remains substantially constant for a predetermined time period (block **530**). In some embodiments, force measurement logic implemented by the circuitry, PCB, or processor requires at least a threshold delay period of quiet time before recording a force reading and converting this to a UO volume measurement. For example, suppose the force sensor senses a possibly transient mechanical impulse from the urine collecting into the bag **120**. In fact, such a measured impulse could be caused by new urine arriving in the bag **120**, by sloshing motion of the existing urine, or by some other transient noise source, and it may be difficult for the sensor to immediately differentiate among these scenarios.

Accordingly, logic implemented by the force sensors and/or circuitry, PCB, or a processor associated with the urine collection bag can be instructed to wait for the sensed force to stabilize, and to remain substantially constant for a threshold period (e.g., 5 or 10 seconds), before recognizing the force as a bona-fide, non-transient signal.

(60) In some embodiments, the system can record and/or transmit the UO measurement to an EMR system in response to an event, such as a detected change in the UO measurement. Accordingly, if the system detects a new force, and subsequently determines that the new force remains substantially constant for at least the threshold time period, it can record and/or transmit the UO measurement.

(61) Having determined that the detected force is not noise and not transient, the system can measure the force using force sensors (block **540**). In various embodiments, the system's force sensors may include one or more strain gauge sensors, pressure sensors, weight sensors, resistive elements, and the like, and are not limited by the present disclosure.

(62) In some embodiments, a network of multiple force sensors can produce multiple force and/or pressure measurements that can be combined, either by simple averaging or by a more sophisticated algorithm or machine learning (ML) method. For example, the system may determine which sensor readings are most reliable at any given time, e.g., as the collection bag tilts, moves, and/or deforms during usage, and weight the individual sensor readings accordingly.

(63) The force sensors may comprise resistive elements with electrical resistance that changes in response to force. In particular, the force sensors may be strain gauge sensors, for example comprising piezoresistors or metallic foil elements that deform in response to strain, thereby changing the electrical resistance of the sensors. The change in the electrical resistance of the sensors may be measured by a printed circuit board (PCB), for example using a Wheatstone bridge, thereby measuring the force on the sensors. In some embodiments, the sensors and the PCB may be powered by a battery and/or a rechargeable battery, so that the urine collection system is portable for a patient's convenient usage. The force sensors may be precise, for example they may be able to measure to within ±2 grams or ±0.02 Newtons, the accuracy needed for treating critical patients. Example force sensors include Model 1075 by Adafruit, Sensor-Puck by Silicon Labs, Model FSR06BE by Ohmite, and FSR 400 Series Force-Sensing Resistors by Interlink Electronics.

(64) A countervailing force may be necessary to stabilize the force sensors, so that they do not accelerate or move, and can therefore measure the strain or pressure accurately. This force may be provided by various sources. For example, the side walls of the collection bag may themselves provide countervailing force to the pressure or strain from the urine, for example if side walls are massive or stiff enough. In a related example, the force sensors may be contained within an internal layer of the side walls and/or the collection bag (see FIG. **2**C). In the latter case, the exterior layer of the side walls and/or the collection bag may provide the countervailing force. In another example, the countervailing force may be provided by an external surface, such as the patient's leg, or another body part (see FIGS. **4**A and **4**B). In yet another example, the countervailing force may be provided by a fabric or elastic strap or reusable collection bag cover wrapped around the collection bag (see FIG. **4**A). Alternatively, the urine collection bag may attach to a chair, wheelchair, bed, or a patient's leg, which may provide the countervailing force.

(65) Next, in the case that the force sensors have measured the weight of the urine in the collection bag, the system can convert this weight measurement to a UO volume based on specific gravity (block **550**), for example by dividing the weight by the specific gravity or density of the urine. Alternatively, in the case that the force sensors have measured the pressure of the urine, the system can convert this pressure measurement to a UO volume based on specific gravity (block **560**). In various embodiments, the system may receive the specific gravity via a user interface, or via Bluetooth or Wi-Fi technology from a local device, a local network, or an Internet cloud server. In some embodiments, force measurement logic implemented by the circuitry, PCB, or processor can perform the conversion in blocks **550** and **560**.

(66) In some embodiments, the system can apply Bernoulli's equation to determine the variation of fluid pressure with depth and/or to convert the force measurements to a volume of UO. For example, the circuitry of the urine collection system may divide the weight, or the equivalent mass, by the urine's specific gravity or density to compute a volume.
(67) Finally, the system can optionally transmit the UO data to an electronic medical records (EMR) system of a hospital or other facility (block **570**). For example, the EMR system can include detailed information about individual patients' UO, such as hourly or twice-hourly UO measurements. In various embodiments, the urine collection system can include Bluetooth and/or Wi-Fi technology, for example in the system's PCB and/or circuitry. In various embodiments, the system can use this technology to transmit the patient's measured UO data directly to a server maintaining the EMR or to transmit the UO data to a server over a network, such as a local-area network, a virtual private network, or the Internet. In various embodiments, the system can send updated UO data over a predetermined schedule, such as at regular intervals, or in response to particular events occurring, for example when the measured volume of urine changes by more than a threshold amount. This can save time for clinicians because they do not need to measure patients' UO every hour, or manually update the EMR or other records.
(68) While some particular embodiments have been disclosed herein, and while the particular embodiments have been disclosed in some detail, it is not the intention for the particular embodiments to limit the scope of the concepts provided herein. Additional adaptations and/or modifications can appear to those of ordinary skill in the art, and, in broader aspects, these adaptations and/or modifications are encompassed as well. Accordingly, departures may be made from the particular embodiments disclosed herein without departing from the scope of the concepts provided herein.

Claims

1. A urine collection bag configured to measure an amount of urine, the urine collection bag comprising: a collection area configured to collect the urine, wherein the urine is received via tubing from a catheter; multiple force sensors coupled to or integrated into the urine collection bag, the multiple force sensors configured to determine multiple force measurements of the urine collected within the collection area, the multiple force measurements corresponding to the multiple force sensors; and circuitry configured to determine the amount of the urine based a combination of the multiple force measurements, and a specific gravity of the urine, wherein the combination of the multiple force measurements includes an average of the multiple force measurements.
2. The urine collection bag of claim 1, wherein while determining the amount of the urine, the circuitry is further configured to: determine whether a force measured by the multiple force sensors remains substantially constant for a predetermined time period, and when a change in the force is detected, divide the force by the specific gravity of the urine to determine an increase of a volume of the urine.
3. The urine collection bag of claim 2, wherein the circuitry receives the specific gravity of the urine as an input via a user interface.
4. The urine collection bag of claim 1, wherein each of the multiple force sensors comprises a resistive element operative to change resistance in response to a force, and wherein the resistance is measured by a printed circuit board (PCB).
5. The urine collection bag of claim 4, wherein the PCB further comprises a low-pass filter operative to filter noise.
6. The urine collection bag of claim 1, wherein: the collection area comprises a flexible pouch, and the multiple force sensors measure a force of the urine by measuring an expansive force of the urine on one or more side walls of the flexible pouch.
7. The urine collection bag of claim 6, wherein the multiple force sensors are situated in an internal

layer of the one or more side walls of the flexible pouch.

8. The urine collection bag of claim 1, wherein the catheter includes one or more of: a small female external catheter (FEC) with an opening disposed on a top side, the opening configured to couple to a female anatomical part, a wicking catheter having a wicking area disposed on a top side of the wicking catheter, a finger-mountable catheter having a finger cavity configured to receive a user finger on a bottom side of the finger-mountable catheter, or a male external catheter (MEC).

9. The urine collection bag of claim 8, wherein the catheter is configured to remain on a patient while the patient stands or walks.

10. The urine collection bag of claim 1, further comprising a wireless transmitter configured to transmit a determined amount of the urine to an electronic medical records system.

11. The urine collection bag of claim 1, wherein the circuitry comprises a printed circuit board (PCB) or a processor.

12. The urine collection bag of claim 1, wherein the multiple force sensors are powered by a battery.

13. A urine collection system comprising: a catheter; a urine collection bag configured to measure an amount of urine, the urine collection bag comprising: a collection area configured to collect the urine, wherein the urine is received via flexible drainage tubing from the catheter, multiple force sensors coupled to or integrated into the urine collection bag, the multiple force sensors configured to determine multiple force measurements of the urine collected within the collection area, the multiple force measurements corresponding to each of the multiple force sensors, and circuitry configured to determine: the amount of the urine based on at least one of the multiple force measurements corresponding to one of the multiple force sensors and a specific gravity of the urine, a difference between different force measurements of the multiple force measurements corresponding to different force sensors of the multiple force sensors, and at least one of tilting, movement, or deformation of the urine collection bag based on the difference; and the flexible drainage tubing coupling the catheter to the urine collection bag.

14. The urine collection system of claim 13, wherein the flexible drainage tubing is configured to be secured to a leg of a patient with either: a stabilization device comprising: an adhesive pad comprising one or more adhesive wings; and a retainer configured to stabilize the flexible drainage tubing, or a fabric strap configured to be wrapped around the leg.

15. The urine collection system of claim 13, wherein while determining the amount of the urine, the circuitry is further configured to: determine whether a force measured by the multiple force sensors remains substantially constant for a predetermined time period, and when a change in the force is detected, divide a pressure by the specific gravity of the urine to determine an increase of a volume of the urine.

16. The urine collection system of claim 15, wherein the circuitry receives the specific gravity of the urine as an input via a user interface.

17. The urine collection system of claim 13, wherein each of the multiple force sensors comprises a resistive element operative to change resistance in response to a force, and further comprising a printed circuit board (PCB) configured to measure the resistance.

18. The urine collection system of claim 17, wherein the PCB further comprises a low-pass filter operative to filter noise.

19. The urine collection system of claim 14, wherein: the collection area of the urine collection bag comprises a flexible pouch, and the multiple force sensors measure a force of the urine by measuring an expansive force of the urine on one or more side walls of the flexible pouch.

20. The urine collection system of claim 19, wherein the multiple force sensors are situated in an internal layer of the one or more side walls of the flexible pouch.

21. The urine collection system of claim 13, wherein the catheter includes one or more of: a small female external catheter (FEC) with an opening disposed on a top side, the opening configured to couple to a female anatomical part, a wicking catheter having a wicking area disposed on a top side

of the wicking catheter, a finger-mountable catheter having a finger cavity configured to receive a user finger on a bottom side of the finger-mountable catheter, or a male external catheter (MEC).
22. The urine collection system of claim 21, wherein the catheter is configured to remain on a patient while the patient stands or walks.
23. The urine collection system of claim 13, further comprising a wireless transmitter configured to transmit a determined amount of the urine to an electronic medical records system.
24. The urine collection system of claim 13, wherein the circuitry comprises a printed circuit board (PCB) or a processor.
25. The urine collection system of claim 13, wherein multiple force sensors are powered by a battery.
